

Descriptive and Frequentist Statistics

Module 3: Descriptive Analysis and Statistical Estimation

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Module Overview

- Foundations: descriptive, inferential and frequentist statistics
- Population, sample, parameter and statistic
- Histogram and distribution analysis
- Mean, variance, standard deviation and standard error
- Skewness and kurtosis
- Order statistics, quantiles, IQR and outliers
- Covariance, correlation and covariance matrices
- PCA and multivariate Gaussian modelling
- Sampling, sampling distributions and Central Limit Theorem
- MSE, bias, variance, unbiasedness and consistency
- Confidence intervals and population proportions
- Parametric estimation: moments and maximum likelihood
- Non-parametric estimation: ECDF and KDE
- Statistical applications in Data Science and AI

1. Introduction to Statistics

- **Statistics** is the science of collecting, organizing, analyzing, interpreting and communicating data.
- It converts observations into quantitative evidence for decision-making.
- Two broad roles:
 1. **Descriptive statistics:** summarize observed data.
 2. **Inferential statistics:** generalize from a sample to a population.
- In AI and Data Science, statistics supports experimentation, feature analysis, uncertainty quantification and model evaluation.

Descriptive, Inferential and Frequentist Statistics

Descriptive statistics

Describes the data actually observed using tables, graphs and numerical summaries.

Inferential statistics

Uses sample information to make statements about an underlying population.

Frequentist statistics

Treats model parameters as fixed but unknown quantities and probability as long-run frequency under repeated sampling.

Important

A confidence interval is a procedure with a long-run coverage property, not a probability statement about a fixed parameter.

Population, Sample, Parameter and Statistic

- **Population:** complete set of units or outcomes of interest.
- **Sample:** subset observed from the population.
- **Parameter:** numerical characteristic of the population.
- **Statistic:** numerical quantity calculated from the sample.

Population $\longrightarrow$ Sampling $\longrightarrow$ Sample $\longrightarrow$ Estimation $\longrightarrow$ Parameter

Examples:

μ = population mean, $\bar{X}$ = sample mean.

Parameter versus Statistic

Aspect	Parameter	Statistic
Refers to	Population	Sample
Typical notation	μ, σ^2, p	$\bar{X}, S^2, \hat{p}$
Usually known?	No	Computable
Role	Target quantity	Estimator/summary

- A statistic may be used as an **estimator** of a parameter.
- Repeated samples produce different statistic values.
- A parameter is fixed for a specified population/model.

Statistics in Data Science and AI

- **EDA**: identify central tendency, spread, skewness and anomalies.
- **Feature engineering**: detect redundant or highly correlated variables.
- **PCA**: reduce dimension using covariance structure.
- **Machine learning**: estimate parameters and quantify prediction error.
- **Experimentation**: compare treatments, models or algorithms.
- **Anomaly detection**: identify observations inconsistent with the data.
- **Uncertainty**: confidence intervals and sampling distributions help communicate reliability.

Example: Statistical Analysis of Features

Suppose a dataset contains student features:

X_1 = study hours, X_2 = attendance, X_3 = previous score.

A statistical workflow is:

1. inspect distributions with histograms;
2. calculate mean, median and standard deviation;
3. calculate skewness and kurtosis;
4. detect outliers using IQR;
5. calculate covariance/correlation;
6. construct the covariance matrix;
7. use PCA if dimensionality reduction is required.

Important

The objective is not merely to calculate numbers, but to understand the structure and uncertainty in the data.

2. Histogram: Definition and Purpose

- A histogram is a graphical representation of the distribution of a quantitative variable.
- The numerical range is divided into **bins** or intervals.
- The height represents frequency, relative frequency or density.
- Unlike a bar chart, histogram bars represent adjacent numerical intervals.

Data → Bins → Frequencies/Densities → Histogram

Uses include detecting center, spread, skewness, modality and unusual values.

Construction of a Histogram

For observations $x_1, \dots, x_n$:

1. Find the minimum and maximum.
2. Choose the number of bins or bin width.
3. Define non-overlapping intervals.
4. Count observations in each interval.
5. Plot the counts/densities against the intervals.

If bin width is h and a bin contains f_j observations, its density can be written as

$$\hat{f}_j = \frac{f_j}{nh}.$$

This makes histogram area approximately sum to one.

Histogram: Frequency and Density

- **Frequency:** number of observations in a bin.
- **Relative frequency:** f_j/n .
- **Density:** relative frequency divided by bin width.

For equal-width bins:

$$\text{density} = \frac{\text{frequency}}{n \times \text{bin width}}.$$

For unequal-width bins, bar **area**, rather than height alone, must represent probability.

Important

A density histogram has total area approximately equal to 1.

Histogram: Worked Example

Consider:

12, 15, 17, 18, 19, 21, 22, 23, 25, 28.

Take bins of width 5:

$[10, 15)$, $[15, 20)$, $[20, 25)$, $[25, 30]$.

Corresponding frequencies:

Bin	Frequency
10 – 15	1
15 – 20	4
20 – 25	3
25 – 30	2

The histogram immediately shows that observations concentrate around the middle intervals.

Interpreting Histogram Shape

Common patterns:

- **Symmetric:** left and right sides are approximately balanced.
- **Right-skewed:** long tail toward larger values.
- **Left-skewed:** long tail toward smaller values.
- **Bimodal:** two prominent peaks.
- **Uniform-like:** relatively even frequencies.
- **Heavy-tailed:** unusually large probability in the tails.

A histogram is exploratory evidence, not proof of a particular distribution.

3. Skewness

Skewness measures the degree and direction of asymmetry in a probability distribution.

The population coefficient of skewness is commonly defined as

$$\gamma_1 = \frac{\mathbb{E}[(X - \mu)^3]}{\sigma^3}$$

For a sample, a simple moment-based estimate is

$$g_1 = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^3}{\left[\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \right]^{3/2}}$$

- $\gamma_1 = 0$: symmetric distribution.
- $\gamma_1 > 0$: **right-skewed**; longer right tail.
- $\gamma_1 < 0$: **left-skewed**; longer left tail.

Important

Skewness is dimensionless and is based on the third central moment.

Skewness: Interpretation and Examples

Type	Skewness	Typical Shape
Symmetric	≈ 0	Balanced tails
Right-skewed	> 0	Long tail to the right
Left-skewed	< 0	Long tail to the left

For a moderately right-skewed distribution, the typical relationship is

$$\text{Mean} > \text{Median} > \text{Mode}$$

For a moderately left-skewed distribution:

$$\text{Mean} < \text{Median} < \text{Mode}$$

Examples:

- Income distributions are often right-skewed.
- Exam scores may be left-skewed when most students score highly.
- Skewness helps determine whether mean-based summaries adequately represent a

4. Kurtosis

Kurtosis describes the heaviness of the tails of a distribution relative to a reference distribution.

The population kurtosis is the standardized fourth central moment:

$$\beta_2 = \frac{\mathbb{E}[(X - \mu)^4]}{\sigma^4}$$

The commonly reported **excess kurtosis** is

$$\gamma_2 = \beta_2 - 3$$

For a normal distribution:

$$\beta_2 = 3, \quad \gamma_2 = 0.$$

- **Mesokurtic**: excess kurtosis ≈ 0 .
- **Leptokurtic**: positive excess kurtosis; heavier tails.
- **Platykurtic**: negative excess kurtosis; lighter tails.

Kurtosis: Interpretation and Comparison

Distribution	Excess Kurtosis	Interpretation
Mesokurtic	≈ 0	Similar tail weight to normal
Leptokurtic	> 0	Heavier tails; more extreme values
Platykurtic	< 0	Lighter tails; fewer extreme values

Kurtosis is based on the fourth central moment:

$$\gamma_2 = \frac{\mathbb{E}[(X - \mu)^4]}{\sigma^4} - 3.$$

- The fourth power gives greater weight to observations far from the mean.
- Therefore, extreme observations can strongly affect kurtosis.
- Kurtosis does **not** simply measure the height or sharpness of the central peak.

Important

Histogram: Bin Width, Advantages and Limitations

- Small bins reveal detail but can create a noisy appearance.
- Large bins smooth the distribution but can hide structure.
- Common choices include Sturges' rule, Scott's rule and Freedman–Diaconis rule.
- Advantages: simple, intuitive and useful for EDA.
- Limitations: appearance depends strongly on bin choice; not ideal for very small samples or precise density estimation.

$$h_{\text{FD}} = 2 \frac{\text{IQR}(X)}{n^{1/3}}.$$

5. Sample Mean

For a sample $x_1, \dots, x_n$:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

Properties under random sampling:

$$\mathbb{E}[\bar{X}] = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

The mean uses every observation and is sensitive to extreme values. It is a fundamental estimator of the population mean.

Sample Mean as an Optimization Solution

The sample mean minimizes the sum of squared deviations:

$$\hat{a} = \arg \min_a \sum_{i=1}^n (x_i - a)^2.$$

Differentiate:

$$\frac{d}{da} \sum_i (x_i - a)^2 = -2 \sum_i (x_i - a).$$

Setting this equal to zero:

$$\sum_i x_i - na = 0 \quad \Rightarrow \quad \boxed{a = \bar{x}}.$$

Thus the mean is the least-squares center of the observations.

Properties of the Sample Mean

- **Linearity:** $\overline{aX + b} = a\bar{X} + b$.
- **Unbiasedness:** $\mathbb{E}[\bar{X}] = \mu$.
- **Consistency:** $\bar{X} \rightarrow \mu$ as $n \rightarrow \infty$ under standard conditions.
- **Low variance:** variance decreases as sample size increases.
- **Sensitivity:** a single extreme observation can substantially change $\bar{X}$.

For independent observations:

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

Variance of the Sample Mean and Standard Error

For independent, identically distributed observations:

$$\text{Var}(\bar{X}) = \text{Var}\left(\frac{1}{n} \sum_i X_i\right) = \frac{\sigma^2}{n}.$$

Therefore,

$$\boxed{SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}}.$$

If σ is unknown, estimate it using S :

$$\widehat{SE}(\bar{X}) = \frac{S}{\sqrt{n}}.$$

Doubling n does not halve the standard error; it reduces it by a factor of $\sqrt{2}$.

Effect of Sample Size on Standard Error

n	$SE(\bar{X})/\sigma$
25	0.20
100	0.10
400	0.05
900	0.0333

$$SE \propto n^{-1/2}.$$

- Larger samples give more stable estimates.
- Precision improves with diminishing returns.
- To reduce standard error by half, approximately four times as many observations are required.

Sample Variance and Standard Deviation

The unbiased sample variance is

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Sample standard deviation:

$$S = \sqrt{S^2}.$$

Interpretation:

- Variance measures average squared deviation from the sample mean.
- Standard deviation is in the same units as the original variable.
- Both measure dispersion around the center.

Why Does Sample Variance Use $n - 1$?

The deviations from the sample mean satisfy

$$\sum_{i=1}^n (X_i - \bar{X}) = 0.$$

Once $n - 1$ deviations are known, the final deviation is determined. Hence there are only $n - 1$ degrees of freedom.

$$\frac{1}{n} \sum_i (X_i - \bar{X})^2$$

tends to underestimate σ^2 . Using Bessel's correction:

$$\boxed{\mathbb{E}[S^2] = \sigma^2}.$$

Thus S^2 is unbiased for population variance.

Sample Variance: Numerical Example

Take:

$$2, 4, 6, 8.$$

Then

$$\bar{x} = 5.$$

Squared deviations:

$$9, 1, 1, 9.$$

Hence

$$\sum (x_i - \bar{x})^2 = 20$$

and

$$S^2 = \frac{20}{4 - 1} = \frac{20}{3} \approx 6.67.$$

Therefore

$$S \approx 2.58.$$

This illustrates the distinction between dividing by n and $n - 1$.

6. Order Statistics

Given a sample $X_1, \dots, X_n$, sort it:

$$X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}.$$

These are called **order statistics**.

- $X_{(1)}$: minimum
- $X_{(n)}$: maximum
- Middle order statistics: basis for medians and quantiles

Order statistics depend on the ranks of observations rather than their original order.

Distribution of an Order Statistic

If $X_1, \dots, X_n$ are i.i.d. with CDF $F(x)$, then for the minimum:

$$P(X_{(1)} > x) = [1 - F(x)]^n.$$

Therefore:

$$F_{X_{(1)}}(x) = 1 - [1 - F(x)]^n.$$

For the maximum:

$$F_{X_{(n)}}(x) = F(x)^n.$$

For a general order statistic $X_{(k)}$, the CDF is

$$P(X_{(k)} \leq x) = \sum_{j=k}^n \binom{n}{j} F(x)^j [1 - F(x)]^{n-j}.$$

Median, Quantiles and Interquartile Range

- The **median** is the 50th percentile.
- Q_1 : 25th percentile.
- Q_3 : 75th percentile.
- Interquartile range:

$$IQR = Q_3 - Q_1.$$

The IQR measures the spread of the central 50% of observations and is more robust to extreme values than variance.

Outlier Detection Using IQR

The standard Tukey rule identifies potential outliers using:

$$\text{Lower fence} = Q_1 - 1.5(IQR)$$

$$\text{Upper fence} = Q_3 + 1.5(IQR).$$

Observations outside these limits are flagged as potential outliers.

- An outlier is not automatically an error.
- It may represent a genuine rare event.
- Domain knowledge is required before removing observations.

7. Sample Covariance

For paired observations (X_i, Y_i) :

$$S_{XY} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}).$$

Interpretation:

- $S_{XY} > 0$: variables tend to increase together.
- $S_{XY} < 0$: one tends to increase as the other decreases.
- $S_{XY} \approx 0$: little linear co-movement.

Covariance: Interpretation and Independence

Covariance depends on the measurement scales:

$$\text{Cov}(aX, bY) = ab \text{ Cov}(X, Y).$$

If X and Y are independent and have finite expectations:

$$\boxed{\text{Cov}(X, Y) = 0}.$$

However,

$$\text{Cov}(X, Y) = 0 \not\Rightarrow X \text{ and } Y \text{ are independent}$$

in general. For jointly Gaussian variables, zero covariance does imply independence.

Covariance: Numerical Example

Consider:

$$X = (1, 2, 3), \quad Y = (2, 4, 6).$$

Then

$$\bar{X} = 2, \quad \bar{Y} = 4.$$

Products of centered values:

$$(-1)(-2), \quad (0)(0), \quad (1)(2) = 2, 0, 2.$$

Thus

$$S_{XY} = \frac{4}{3-1} = 2.$$

The positive value reflects perfect positive linear association in this example.

Covariance and Correlation

Pearson sample correlation:

$$r = \frac{S_{XY}}{S_X S_Y}.$$

Properties:

$$-1 \leq r \leq 1.$$

- $r = 1$: perfect positive linear relation.
- $r = -1$: perfect negative linear relation.
- $r \approx 0$: weak linear association.

Correlation is dimensionless; covariance is not.

Important

Correlation measures linear association, not causation.

Sample Covariance Matrix

For a p -dimensional observation vector

$$\mathbf{x}_i = (X_{i1}, \dots, X_{ip})^T,$$

the sample covariance matrix is

$$S = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T.$$

It has the form

$$S = \begin{bmatrix} S_{11} & S_{12} & \cdots \\ S_{21} & S_{22} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}.$$

Diagonal entries are variances; off-diagonal entries are covariances.

Properties of the Covariance Matrix

A covariance matrix:

- is square and symmetric: $S = S^T$;
- has non-negative diagonal elements;
- is positive semidefinite:

$$\mathbf{a}^T S \mathbf{a} \geq 0;$$

- describes scale and dependence among variables;
- is central to PCA, Gaussian models and multivariate analysis.

If variables are standardized, the covariance matrix becomes the correlation matrix.

Covariance Matrix: Example

For two variables:

$$S = \begin{bmatrix} 4 & 2 \\ 2 & 9 \end{bmatrix}.$$

Interpretation:

$$\text{Var}(X_1) = 4, \quad \text{Var}(X_2) = 9, \quad \text{Cov}(X_1, X_2) = 2.$$

The corresponding correlation is

$$r = \frac{2}{\sqrt{4}\sqrt{9}} = \frac{1}{3}.$$

Thus the variables have positive but moderate linear association.

8. Covariance Matrix and PCA

Principal Component Analysis (PCA) finds directions of maximum variance. For covariance matrix S :

$$S\mathbf{v}_j = \lambda_j\mathbf{v}_j.$$

Here λ_j is the variance along eigenvector $\mathbf{v}_j$. Order:

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p \geq 0.$$

The first principal component is the direction associated with the largest eigenvalue.

Variance Explained in PCA

The proportion of variance explained by component j is

$$\frac{\lambda_j}{\sum_{k=1}^p \lambda_k}.$$

Cumulative explained variance:

$$\frac{\sum_{j=1}^m \lambda_j}{\sum_{k=1}^p \lambda_k}.$$

PCA can therefore select the smallest m retaining a desired percentage of total variance, such as 90% or 95%.

Multivariate Gaussian Model

A multivariate normal distribution is written

$$\mathbf{X} \sim \mathcal{N}_p(\boldsymbol{\mu}, \Sigma).$$

Its density is

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp \left[-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right].$$

- $\boldsymbol{\mu}$: mean vector
- Σ : covariance matrix
- Σ controls spread and orientation of the distribution.

9. Frequentist Statistics and Random Sampling

Frequentist inference is based on repeated sampling from a population/model.

- The parameter is fixed but unknown.
- The sample is random.
- Consequently, an estimator is a random variable.
- Its probability distribution is the **sampling distribution**.

A good sampling design attempts to make the sample representative of the target population.

Sampling Methods

Major probability sampling methods:

- **Simple random sampling:** every unit has equal selection probability.
- **Systematic sampling:** select every k th unit after a random start.
- **Stratified sampling:** divide into strata and sample within each.
- **Cluster sampling:** sample naturally occurring groups.

Non-probability methods include convenience and voluntary-response sampling, which can create substantial bias.

Sampling Bias

Sampling bias occurs when the sampling mechanism systematically over- or under-represents parts of the target population. Examples:

- coverage bias;
- non-response bias;
- voluntary-response bias;
- convenience sampling;
- survivorship bias.

Important

Increasing sample size does not automatically remove systematic bias.

A very large biased sample can still give a biased estimate.

Sampling Distribution of the Sample Mean

Suppose

$$X_1, \dots, X_n \stackrel{iid}{\sim} (\mu, \sigma^2).$$

Then

$$\mathbb{E}[\bar{X}] = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

If the population is normal:

$$\boxed{\bar{X} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)}.$$

Even when the original population is not normal, the distribution of $\bar{X}$ often becomes approximately normal for sufficiently large n .

Central Limit Theorem

Under standard regularity conditions:

$$\boxed{\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1)} \quad (n \rightarrow \infty).$$

Meaning:

- independent observations with finite mean and variance are typical assumptions;
- the standardized sample mean approaches a standard normal distribution;
- exact normality of the population is not required.

The CLT is one of the foundations of large-sample inference.

Importance of the Central Limit Theorem

The CLT enables:

- approximate confidence intervals;
- hypothesis tests;
- normal-based error approximations;
- statistical quality control;
- scalable inference for many real-world datasets.

Population distribution $\longrightarrow$ Sampling distribution of $\bar{X} \approx$ Normal for large n .

The CLT concerns sampling distributions, not the claim that the raw data themselves become normal.

10. Mean Square Error

For estimator $\hat{\theta}$ of parameter θ :

$$MSE(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2].$$

MSE combines two ideas:

$$\text{accuracy} = \text{bias} + \text{variability}.$$

It is widely used for comparing estimators and evaluating regression models. Smaller MSE generally indicates better expected squared accuracy for the same target parameter.

Bias of an Estimator

Bias is

$$\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta.$$

An estimator is unbiased if

$$\mathbb{E}[\hat{\theta}] = \theta.$$

Important:

- Bias is a systematic tendency away from the target.
- An unbiased estimator can still have high variance.
- A biased estimator can sometimes have lower MSE.

Therefore, unbiasedness alone is not a complete criterion for estimator quality.

Bias–Variance Decomposition

For squared error:

$$MSE(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2.$$

For a prediction model, expected test error can similarly be decomposed conceptually into:

irreducible noise + bias² + variance.

- High bias → underfitting tendency.
- High variance → overfitting tendency.
- Model complexity often trades one against the other.

MSE in Machine Learning: Numerical Example

Suppose an estimator has

$$\text{Bias}(\hat{\theta}) = 2, \quad \text{Var}(\hat{\theta}) = 9.$$

Then

$$MSE = 9 + 2^2 = 13.$$

If another estimator has

$$\text{Bias} = 1, \quad \text{Var} = 5,$$

then

$$MSE = 5 + 1 = 6.$$

The second estimator is preferable under squared-error loss even though bias is not zero.

11. Consistency of Estimators

An estimator $\hat{\theta}_n$ is **consistent** for θ if

$$\boxed{\hat{\theta}_n \xrightarrow{p} \theta} \quad (n \rightarrow \infty).$$

That is, for every $\epsilon > 0$,

$$P(|\hat{\theta}_n - \theta| > \epsilon) \rightarrow 0.$$

Consistency describes asymptotic behavior as sample size grows. It is different from finite-sample unbiasedness.

Consistency of the Sample Mean

For i.i.d. observations with finite variance:

$$\mathbb{E}[\bar{X}] = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n} \rightarrow 0.$$

By Chebyshev's inequality:

$$P(|\bar{X} - \mu| \geq \epsilon) \leq \frac{\sigma^2}{n\epsilon^2} \rightarrow 0.$$

Therefore:

$$\boxed{\bar{X} \xrightarrow{p} \mu}.$$

This is the weak law of large numbers in action.

Unbiasedness versus Consistency

	Unbiasedness	Consistency
Nature	Finite-sample property	Asymptotic property
Condition	$\mathbb{E}[\hat{\theta}] = \theta$	$\hat{\theta}_n \rightarrow_p \theta$
Focus	Average target correctness	Large- n convergence
Can coexist?	Yes	

Important

An estimator can be biased but consistent. An estimator can be unbiased without having especially small finite-sample variance.

12. Confidence Intervals

A confidence interval is a random interval constructed from data:

$$[L(X), U(X)].$$

For a 95% confidence procedure:

$$P_{\theta}\{L(X) \leq \theta \leq U(X)\} = 0.95$$

under the assumed model and repeated sampling. The parameter θ is fixed; the interval endpoints vary from sample to sample.

Confidence Interval with Known σ

If the population standard deviation is known:

$$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}.$$

For 95% confidence:

$$z_{0.025} \approx 1.96.$$

Hence:

$$\bar{X} \pm 1.96 \frac{\sigma}{\sqrt{n}}.$$

The margin of error is

$$E = z_{\alpha/2} \frac{\sigma}{\sqrt{n}}.$$

Confidence Interval with Unknown σ

When σ is unknown, use S :

$$\bar{X} \pm t_{\alpha/2, n-1} \frac{S}{\sqrt{n}}.$$

The t distribution has heavier tails than the standard normal for finite degrees of freedom. As n increases:

$$t_{\alpha/2, n-1} \rightarrow z_{\alpha/2}.$$

This is the standard one-sample t interval for a population mean.

Confidence Interval: Worked Example

Suppose

$$n = 25, \quad \bar{x} = 72, \quad S = 10.$$

For a 95% interval:

$$t_{0.025, 24} \approx 2.064.$$

Thus

$$72 \pm 2.064 \left(\frac{10}{5} \right) = 72 \pm 4.128.$$

Therefore:

$$\boxed{(67.872, 76.128)}.$$

The interval is a plausible range generated by the specified frequentist procedure for the population mean.

Correct Frequentist Interpretation

A 95% confidence procedure means:

$$P(L(X) \leq \theta \leq U(X)) = 0.95$$

before observing the sample, under repeated sampling. Correct wording:

“If this procedure were repeated many times under the model, about 95% of the resulting intervals would contain θ .”

Avoid saying that there is a 95% probability that the fixed parameter lies inside this already-observed interval.

Confidence Interval Width and Population Proportion

For a mean:

$$\text{Width} \propto \frac{\text{critical value} \times \text{standard deviation}}{\sqrt{n}}.$$

Width increases with variability and confidence level, and decreases with n . For a population proportion:

$$\hat{p} = \frac{x}{n}, \quad SE(\hat{p}) \approx \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}.$$

A large-sample Wald interval is

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}.$$

Other intervals, such as Wilson intervals, can have better finite-sample behavior.

13. Parametric Model Estimation

A parametric model assumes a finite-dimensional parameter:

$$X \sim f(x; \theta), \quad \theta \in \Theta \subseteq \mathbb{R}^k.$$

Examples:

- Normal: $\theta = (\mu, \sigma^2)$
- Bernoulli: $\theta = p$
- Poisson: $\theta = \lambda$

Estimation uses sample data to infer θ . Parametric models can be statistically efficient when their assumptions are reasonable.

Method of Moments

The method of moments equates sample moments to theoretical moments. For example:

$$\mathbb{E}[X] = \mu \quad \Rightarrow \quad \bar{X} = \mu.$$

For a distribution with parameters $\theta_1, \dots, \theta_k$:

$$m'_j = \frac{1}{n} \sum_{i=1}^n X_i^j$$

is matched with

$$\mathbb{E}[X^j].$$

Solving the resulting equations produces method-of-moments estimators.

Maximum Likelihood Estimation

For observations $x_1, \dots, x_n$ with density/PMF $f(x; \theta)$:

$$L(\theta) = \prod_{i=1}^n f(x_i; \theta).$$

The maximum likelihood estimator is

$$\hat{\theta}_{MLE} = \arg \max_{\theta \in \Theta} L(\theta).$$

Since products can be inconvenient, maximize the log-likelihood:

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^n \log f(x_i; \theta).$$

The logarithm is monotone, so it does not change the maximizing parameter.

MLE of Normal Mean and Variance

If

$$X_i \sim \mathcal{N}(\mu, \sigma^2),$$

the log-likelihood is

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_i (X_i - \mu)^2.$$

Differentiation gives:

$$\hat{\mu}_{MLE} = \bar{X}$$

and

$$\hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum_i (X_i - \bar{X})^2.$$

Notice the variance denominator is n , not $n - 1$.

MLE versus Unbiased Sample Variance

Two estimators:

$$\hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum_i (X_i - \bar{X})^2$$

and

$$S^2 = \frac{1}{n-1} \sum_i (X_i - \bar{X})^2.$$

For normal data:

$$\mathbb{E}[\hat{\sigma}_{MLE}^2] = \frac{n-1}{n} \sigma^2.$$

Thus the MLE is biased downward in finite samples. But it is the likelihood maximizer, whereas S^2 is unbiased. This illustrates that different estimator criteria can lead to different choices.

14. Non-Parametric Model Estimation

Non-parametric methods impose fewer assumptions about the functional form of the population distribution. Examples:

- Empirical Distribution Function (ECDF)
- Kernel Density Estimation (KDE)
- Rank-based methods
- Quantile-based summaries

Advantages:

- flexible;
- useful when distributional assumptions are uncertain.

Trade-off: greater flexibility can require more data and careful tuning.

Empirical Distribution Function

For observations $X_1, \dots, X_n$:

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n I(X_i \leq x).$$

Properties:

- non-decreasing;
- right-continuous;
- ranges from 0 to 1;
- jumps by $1/n$ at each observation, accounting for ties.

The ECDF estimates the population CDF directly without specifying a parametric family.

Kernel Density Estimation

KDE estimates an unknown density using:

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - X_i}{h}\right).$$

Here:

- K : kernel function;
- $h > 0$: bandwidth;
- n : sample size.

Common kernels include Gaussian, Epanechnikov, uniform and triangular kernels. The bandwidth generally has a larger influence on the result than the exact kernel choice.

KDE: Normalization, Bandwidth and Bias–Variance Trade-off

If

$$\int K(u) du = 1,$$

then

$$\int \hat{f}_h(x) dx = 1.$$

Bandwidth:

- small h : detailed, wiggly estimate, potentially high variance;
- large h : smooth estimate, potentially high bias.

Thus KDE exhibits a bias–variance trade-off. Bandwidth can be selected using rules of thumb, cross-validation or other data-driven criteria.

15. Parametric versus Non-Parametric Estimation

	Parametric	Non-parametric
Assumptions	Specific distributional form	Fewer shape assumptions
Flexibility	Lower	Higher
Data requirement	Often lower	Often higher
Interpretability	Often high	Depends on method
Examples	Normal MLE, Poisson	ECDF, KDE

16. Integrated Statistical Analysis Workflow

A practical workflow:

Collect → Clean → Explore → Summarize → Model → Infer → Validate

- Histograms reveal distributional shape.
- Mean/median describe central tendency.
- Variance/IQR describe dispersion.
- Skewness describes asymmetry.
- Kurtosis describes tail behavior.
- Covariance/correlation describe relationships.
- Confidence intervals quantify sampling uncertainty.
- Estimation methods connect data to population/model parameters.

17. Applications in Data Science and AI

- **Exploratory Data Analysis:** distributions, skewness, kurtosis, outliers and feature relationships.
- **PCA:** covariance matrix and eigenvalue-based dimensionality reduction.
- **Regression:** least squares, residual variance and prediction error.
- **Anomaly detection:** unusual values relative to estimated distributions.
- **Classification:** probabilistic models and uncertainty.
- **A/B testing:** sampling distributions, confidence intervals and inference.
- **Scientific/engineering systems:** measurement uncertainty, quality control and reliability analysis.

18. Key Comparisons and Formula Sheet

Concept	Key expression
Sample mean	$\bar{X} = \frac{1}{n} \sum X_i$
Sample variance	$S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$
Standard error	$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$
Skewness	$\gamma_1 = \frac{\mathbb{E}[(X - \mu)^3]}{\sigma^3}$
Kurtosis	$\beta_2 = \frac{\mathbb{E}[(X - \mu)^4]}{\sigma^4}$
Excess kurtosis	$\gamma_2 = \beta_2 - 3$
Covariance	$S_{XY} = \frac{1}{n-1} \sum (X_i - \bar{X})(Y_i - \bar{Y})$
Correlation	$r = S_{XY} / (S_X S_Y)$
MSE	$\text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2$
Consistency	$\hat{\theta}_n \xrightarrow{P} \theta$
Known- σ CI	$\bar{X} \pm z_{\alpha/2} \sigma / \sqrt{n}$
Unknown- σ CI	$\bar{X} \pm t_{\alpha/2, n-1} S / \sqrt{n}$

References

1. George Casella and Roger L. Berger, *Statistical Inference*, Cengage Learning.
2. Larry Wasserman, *All of Statistics*, Springer.
3. Trevor Hastie, Robert Tibshirani and Jerome Friedman, *The Elements of Statistical Learning*, Springer.
4. Gareth James et al., *An Introduction to Statistical Learning*, Springer.
5. Christopher M. Bishop, *Pattern Recognition and Machine Learning*, Springer.
6. Sheldon Ross, *A First Course in Probability*, Pearson.